



AI-Powered Predictive Maintenance for Industrial Equipment

Milestone 2: Advanced Data Analysis & Feature Engineering

Executive Summary

This report documents the completion of Milestone 2, focused on enriching the dataset for high-accuracy predictive modeling. The key achievements include establishing temporal patterns through decomposition, identifying instability via anomaly detection, and creating degradation-aware features (rolling statistics and rate of change). The final step, PCA, validated that **Temperature** and **Vibration** are the primary drivers of data variance, confirming the dataset is optimized for failure prediction in Milestone 3.

Table of Contents

Section	Topic	Page (Reference)
I	Introduction & Dataset Overview	3
II	Advanced Data Analysis	4
	2.1 Time-Series Analysis & Trends	
	2.2 Anomaly Detection for Instability	
III	Feature Engineering Enhancements	6
	3.1 Rate of Change Features	
	3.2 Degradation Patterns (Rolling Statistics)	
IV	Dimensionality Reduction & Failure Indicators	7
	4.1 Principal Component Analysis (PCA) Findings	
	4.2 Key Failure Indicators Assessment	
V	Data Visualization & Key Takeaways	9

I. Introduction & Dataset Overview

Objective	Rationale for Inclusion
Enhance Feature Extraction	Raw sensor readings are often noisy and lack context. Extraction creates features (e.g., trend, rate) that are directly interpretable by ML models as signs of degradation.
Perform Advanced Analysis	Methods like PCA identify the fundamental structure of the data, isolating latent factors (combinations of sensors) that drive the most variance and are most predictive of failure.

Dataset Context

The analysis was performed on 100,000 minute-by-minute records from 50 unique machines. The target variable, `maintenance_required` (19.7% incidence), confirms a significant class imbalance that must be addressed in the subsequent modeling phase.

Feature	Data Type	Key Metric	Significance
timestamp	Datetime	1-minute intervals	Crucial for ordering data and generating time-dependent features.
temperature	Float64	Mean: 75.02 °C	Primary indicator of thermal load and friction.
vibration	Float64	Mean: 50.01	Primary indicator of mechanical wear and instability.

II. Advanced Data Analysis

This section details the methods used to interpret the temporal structure of the sensor data.

• 2.1 Time-Series Analysis & Trends

Task	Methodology & Tool	Rationale
Identify Structural Patterns	Time-Series Decomposition (Additive Model, statsmodels.tsa) applied to the temperature sensor.	To separate the signal into its components (Trend, Seasonality, Residual) and prove whether the data exhibits predictable, non-random behavior.

Findings:

- **Trend:** A slow, positive trend was confirmed, signifying cumulative wear and long-term degradation.
- **Seasonality:** A strong 24-hour cycle was observed, confirming daily operational patterns (e.g., shifts or energy fluctuations) influence sensor readings.

2.2 Anomaly Detection for Instability

Task	Methodology & Threshold	Rationale
Detect Sudden Instabilities	Z-score Test applied to vibration and temperature. A threshold of **	Z-score

Output: Two new binary features, `vibration_zscore_anomaly` and `temp_zscore_anomaly`, were created.

Insight: These anomaly flags serve as crucial, high-impact binary features in the model, capturing the non-gradual, abrupt failures missed by trend analysis.

III. Feature Engineering Enhancements

The objective was to develop new features that translate the complex concept of machine deterioration into quantifiable variables.

Engineered Feature Category	New Features	Rationale and Expected Impact
Rate of Change	temp_diff, vibration_diff, pressure_diff	Rationale: Measures the <i>instantaneous</i> rate of change. Rapid acceleration in sensor readings (high diff values) is often a strong indicator of an impending mechanical failure (e.g., bearing overheating rapidly).
Degradation Patterns	temp_roll_mean, vibration_roll_mean	Methodology: Calculated using a 30-minute rolling window grouped by machine_id. Rationale: Smooths short-term noise and captures the medium-term accumulation of wear which signals system degradation before a critical event occurs.
Temporal Context	elapsed_time	Methodology: Calculates the total seconds passed since the machine's first log. Rationale: Provides the model with a measure of the machine's cumulative operational age or total runtime, which is crucial for predicting Remaining Useful Life (RUL).

Implementation Snippet (Conceptual):

Python

```
# Capturing Degradation Pattern
df['temp_roll_mean'] = df.groupby('machine_id')['temperature'].transform(
    lambda x: x.rolling(window=30).mean()
)

# Capturing Rate of Change
df['vibration_diff'] = df.groupby('machine_id')['vibration'].transform(
    lambda x: x.diff()
)
```

IV. Dimensionality Reduction & Failure Indicators

• 4.1 Principal Component Analysis (PCA) Findings

Methodology: PCA was applied to the five standardized raw sensor features.

Rationale: PCA identifies the fundamental, independent factors that underpin the data. This validates that the sensors we are focusing on (**Temperature** and **Vibration**) are indeed the primary sources of information and predictive power.

Principal Component	Explained Variance	Cumulative Variance	Interpretation based on Loadings
PC1	%48.0	%48.0	Thermal/Load State : Heavily driven by high correlation (loadings) with Temperature and Energy Consumption ,reflecting the machine's primary operating load.
PC2	%23.0	%71.0	Mechanical/Environmental Stress : Defined by Vibration (positive correlation) contrasting sharply with Humidity (negative correlation), reflecting instability under different environmental conditions.

PCA Conclusion: With 71% of the data's complexity captured by two components, the analysis confirms that predictive effort should be concentrated on features related to thermal load and mechanical instability.

• 4.2 Key Failure Indicators Assessment

The mean values of raw sensors were compared between maintenance-required and non-maintenance groups.

Feature	Maintenance Required (Mean)	No Maintenance (Mean)	Difference (Required - No Maint.)
temperature	80.748	73.609	+7.139
vibration	53.434	49.173	+4.260
pressure	2.996	3.002	-0.006

Result: The significant difference in means confirmed **Temperature** and **Vibration** as the **primary critical failure indicators**, justifying the extensive feature engineering performed on them.

V. Data Visualization & Key Takeaways

Data Visualization: Correlation Analysis:

A correlation heatmap was generated to confirm the linear relationships between all features.

Findings:

- **Validation of Engineered Features:** The target variable showed the strongest association with the rolling mean features, proving that the engineered degradation patterns are more predictive than the raw sensor readings.
- **Operational Context:** The strong correlation between Temperature and Energy Consumption validated the link between high operational load and increased thermal stress.

Conclusion and Key Takeaways

The activities in Milestone 2 successfully transformed the raw dataset into a highly predictive feature set, ready for efficient Machine Learning modeling.

Milestone 2 successfully built a robust predictive foundation for the project.

- 1. Modeling Mechanism Established:** The feature set now includes both indicators of gradual degradation (rolling mean) and immediate instability (diff and anomalies).
- 2. PCA Validation:** PCA confirmed that the focus on thermal and mechanical indicators is statistically sound, as these factors explain the vast majority of data variance.
- 3. Data Readiness:** The dataset is fully transformed, incorporating all necessary temporal and statistical context, and is now ready for the development and optimization of machine learning models in Milestone3

Readiness for Milestone 3

Challenge Addressed in Milestone 2	Impact on Modeling (Milestone 3)
Noisy Data and Instability	Data is smoothed via rolling means, leading to more stable model training.
Lack of Temporal Context	Features like diff and roll_mean provide crucial sequential information for time-series models.
Dimensionality Reduction	PCA components can be used to speed up training and mitigate the risk of overfitting.

Conclusion: The dataset is now fully engineered, containing indicators of **cumulative degradation** and **instantaneous instability**, making it robust and prepared for the model development and optimization phase.